COMPREHENSIVE MEDICAL REFERENCE GUIDE

Diseases · Symptoms · Prevention · Treatment · Medical Insights

Curated Knowledge Base for Healthcare RAG Systems

12+	Disease Categories	50+	Conditions Covered	200+	Symptoms Listed	RAG Ready	Structured Content

DISCLAIMER: This document is compiled for AI/RAG knowledge-base purposes. It does not constitute professional medical advice. Always consult a qualified healthcare provider for diagnosis and treatment.

CHAPTER —

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CHAPTER 1

Infectious & Communicable Diseases

Pathogens · Transmission · Symptoms · Treatment · Prevention

1.1 Influenza (Flu)

Influenza is an acute viral respiratory illness caused by influenza A, B, or C viruses. It spreads rapidly via respiratory droplets and remains one of the leading causes of seasonal morbidity worldwide.

Symptoms	Risk Factors
• Sudden high fever (38–40 °C) • Severe headache & myalgia • Non-productive dry cough • Sore throat & rhinorrhoea • Fatigue and malaise • Chills and rigors • Occasional vomiting/diarrhoea (children)	• Age >65 or <5 years • Pregnancy • Chronic lung/heart/kidney disease • Immunocompromised states • Obesity (BMI >40) • Healthcare worker exposure
Prevention	**Treatment**
• Annual influenza vaccination (primary measure) • Hand hygiene and respiratory etiquette • Avoid close contact with infected persons • Antiviral chemoprophylaxis (oseltamivir) for high-risk contacts	• Antiviral: Oseltamivir (Tamiflu) within 48 h of onset • Zanamivir (inhaled) — alternative • Baloxavir marboxil — single-dose newer option • Symptomatic: antipyretics, rest, hydration • Hospitalisation for severe/complicated cases

Note: Influenza can cause severe complications including pneumonia, myocarditis, and encephalitis especially in vulnerable populations.

1.2 COVID-19 (SARS-CoV-2)

COVID-19 is caused by the SARS-CoV-2 coronavirus. Declared a pandemic in March 2020, it has caused significant global mortality and long-term sequelae ('Long COVID') in survivors.

Common Symptoms (Acute Phase)

• Fever, chills, and sweats

• Cough (dry or productive)

• Shortness of breath / dyspnoea

• Loss of taste (ageusia) and/or smell (anosmia) — hallmark symptoms

• Fatigue, body aches, headache

- Sore throat, nasal congestion
- Nausea, vomiting, diarrhoea

Severe Disease Indicators (Seek Emergency Care)

- Persistent chest pain or pressure
- Oxygen saturation (SpO_2) <94% on room air
- Confusion or inability to stay awake
- Cyanosis (bluish lips or face)
- Respiratory rate >30 breaths/min

Prevention
- mRNA vaccines (Pfizer-BioNTech, Moderna)
- Adenoviral-vector vaccines (Johnson & Johnson, AstraZeneca)
- Mask-wearing in high-risk settings
- Physical distancing and ventilation
- Hand hygiene — soap ≥20 seconds
- Isolation when symptomatic

Treatment
- Antivirals: Nirmatrelvir/ritonavir (Paxlovid), Remdesivir
- Monoclonal antibodies (select variants)
- Corticosteroids (dexamethasone) for severe/critical cases
- Supplemental O_2, prone positioning, mechanical ventilation
- Anticoagulation for thromboembolism risk

1.3 Tuberculosis (TB)

TB is caused by Mycobacterium tuberculosis, primarily affecting the lungs (pulmonary TB) but capable of infecting any organ (extrapulmonary TB). It is spread via airborne droplet nuclei and remains the world's second leading infectious disease killer after COVID-19.

Symptoms (Pulmonary TB)
- Chronic productive cough >3 weeks
- Haemoptysis (blood in sputum)
- Night sweats
- Unexplained weight loss
- Low-grade fever (afternoon)
- Fatigue and anorexia
- Chest pain & breathlessness

Diagnosis
- Sputum smear microscopy (AFB)
- GeneXpert MTB/RIF (rapid molecular test)
- Culture on Lowenstein-Jensen medium
- Chest X-ray: cavities, infiltrates
- Tuberculin skin test (Mantoux)
- IGRA (Interferon-Gamma Release Assay)
- CT scan for extrapulmonary disease

Treatment — Standard DOTS Regimen

- Intensive phase (2 months): Isoniazid + Rifampicin + Pyrazinamide + Ethambutol (2HRZE)
- Continuation phase (4 months): Isoniazid + Rifampicin (4HR)
- MDR-TB: Bedaquiline, Linezolid, Levofloxacin-based regimens (18–24 months)
- BCG vaccination at birth — primary prevention strategy
- Contact tracing and chemoprophylaxis for latent TB

1.4 Malaria

Malaria is caused by Plasmodium parasites (P. falciparum, P. vivax, P. malariae, P. ovale) transmitted through bites of infected female Anopheles mosquitoes. P. falciparum causes the most severe and fatal form.

Symptoms
- Classic triad: fever, chills, sweating
- Cyclical fever patterns (every 48–72 h)
- Headache and myalgia
- Nausea, vomiting, diarrhoea
- Anaemia and jaundice
- Splenomegaly (enlarged spleen)
- Severe: cerebral malaria, multi-organ failure

Prevention
- Insecticide-treated mosquito nets (ITNs)
- Indoor residual spraying (IRS)
- Chemoprophylaxis (Atovaquone-Proguanil, Doxycycline)
- R21/Matrix-M vaccine (approved 2023) & RTS,S/AS01
- Protective clothing and DEET repellents
- Draining stagnant water (vector control)

Treatment

- Uncomplicated P. falciparum: Artemisinin-based combination therapy (ACT) — e.g., Artemether-Lumefantrine
- Severe malaria: IV Artesunate (preferred) or IV Quinine
- P. vivax/ovale: Chloroquine + Primaquine (to eliminate liver hypnozoites — check G6PD status)
- P. malariae: Chloroquine

1.5 Dengue Fever

Dengue is a mosquito-borne viral infection caused by DENV 1–4 serotypes, transmitted by Aedes aegypti and Aedes albopictus mosquitoes. Second infection with a different serotype increases risk of severe dengue (dengue haemorrhagic fever / dengue shock syndrome).

Warning Signs (Severe Dengue)
- Abdominal pain or tenderness
- Persistent vomiting
- Clinical fluid accumulation
- Mucosal bleeding
- Lethargy / restlessness
- Liver enlargement >2 cm
- Rapid rise in haematocrit with platelet fall

Management
- No specific antiviral; supportive care
- Oral/IV fluid resuscitation (guided by haematocrit)
- Paracetamol for fever (avoid NSAIDs/aspirin — bleeding risk)
- Platelet transfusion if <10,000 or active bleeding
- Dengvaxia vaccine (9–45 yr, seropositive only)
- Mosquito bite prevention strategies

1.6 Viral Hepatitis (A, B, C)

Feature	Hepatitis A	Hepatitis B	Hepatitis C
Pathogen	HAV (RNA virus)	HBV (DNA virus)	HCV (RNA virus)
Transmission	Faecal-oral (water/food)	Blood, sexual, perinatal	Blood/needles (mostly)

Chronicity	No — self-limiting	5–10% adults, 90% neonates	~75% become chronic
Vaccine	Yes (HAV vaccine)	Yes (HBV vaccine)	No vaccine available
Treatment	Supportive care	Antivirals: TDF, ETV, TAF	DAAs: Sofosbuvir + Ledipasvir/Daclatasvir
Complications	Rare, fulminant hepatitis	Cirrhosis, HCC	Cirrhosis, HCC

1.7 HIV/AIDS

Human Immunodeficiency Virus (HIV) attacks CD4+ T-lymphocytes, progressively impairing cell-mediated immunity. Without treatment, HIV leads to Acquired Immunodeficiency Syndrome (AIDS), defined by CD4 count <200 cells/µL or AIDS-defining illness.

Stages & Symptoms

- Acute HIV: flu-like illness (2–4 wk post-exposure)
- Chronic/Latent: may be asymptomatic for years
- AIDS: opportunistic infections (PCP, CMV, toxoplasmosis)
- Constitutional: weight loss, night sweats, lymphadenopathy
- Neurological: AIDS dementia complex

Treatment — ART

- First-line: TDF + 3TC/FTC + DTG (dolutegravir-based)
- Aim: viral load <50 copies/mL (undetectable = untransmittable)
- PrEP: Tenofovir + Emtricitabine for HIV-negative at risk
- PEP: 28-day ART regimen within 72 h of exposure
- Treat all HIV+ persons regardless of CD4 count
- Regular CD4 count & viral load monitoring

CHAPTER 2

Cardiovascular Diseases

Heart · Vessels · Blood Pressure · Stroke

2.1 Coronary Artery Disease (CAD)

CAD results from atherosclerotic plaque formation in coronary arteries, reducing blood flow to the myocardium. It is the leading cause of death globally. Acute manifestations include unstable angina and myocardial infarction (heart attack).

Risk Factors

- Hypertension
- Hypercholesterolaemia (high LDL)
- Smoking (doubles risk)
- Type 2 Diabetes
- Obesity and sedentary lifestyle
- Family history (first-degree relative <55 M / <65 F)
- Chronic kidney disease

Symptoms of Acute MI

- Crushing central chest pain (>20 min)
- Radiation to arm, jaw, back, or shoulder
- Diaphoresis (profuse sweating)
- Nausea, vomiting, shortness of breath
- Sense of impending doom
- Women: atypical — fatigue, jaw pain, nausea
- Silent MI in elderly/diabetics

Treatment of Acute MI (STEMI — Time Critical)

- MONA protocol: Morphine, Oxygen, Nitrates, Aspirin
- Primary PCI (percutaneous coronary intervention) within 90 min — gold standard
- Thrombolysis (streptokinase/tPA) if PCI unavailable within 120 min
- Dual antiplatelet: Aspirin + P2Y12 inhibitor (Ticagrelor/Clopidogrel)
- Anticoagulation: Heparin or Enoxaparin
- Beta-blockers, ACE inhibitors, Statins — long-term secondary prevention

2.2 Hypertension (High Blood Pressure)

Hypertension is defined as sustained blood pressure ≥130/80 mmHg (AHA 2017) or ≥140/90 mmHg (WHO). It is the most prevalent cardiovascular risk factor, affecting ~1.3 billion people globally and often asymptomatic ('silent killer').

Category	Systolic (mmHg)	Diastolic (mmHg)	Action
Normal	<120	<80	Maintain healthy lifestyle
Elevated	120–129	<80	Lifestyle modification
Stage 1 HTN	130–139	80–89	Lifestyle ± single drug

Stage 2 HTN	≥140	≥90	Combination drug therapy
Hypertensive Crisis	≥180	≥120	Emergency care immediately

Lifestyle Modifications
- DASH diet (fruits, vegetables, low sodium)
- Sodium restriction <2.3 g/day
- Regular aerobic exercise (150 min/week)
- Weight loss (5 mmHg per 5 kg reduction)
- Alcohol moderation
- Smoking cessation
- Stress management

Drug Classes
- ACE inhibitors (Enalapril, Ramipril)
- ARBs (Losartan, Valsartan)
- Calcium channel blockers (Amlodipine)
- Thiazide diuretics (Hydrochlorothiazide)
- Beta-blockers (Metoprolol, Atenolol)
- Aldosterone antagonists (Spironolactone)

2.3 Stroke (Cerebrovascular Accident)

A stroke occurs when blood supply to part of the brain is interrupted (ischaemic, 85%) or a blood vessel ruptures (haemorrhagic, 15%). Time-critical: 'Time is Brain' — 1.9 million neurons die every minute without treatment.

FAST / BE-FAST Warning Signs

- B — Balance: sudden loss of balance or coordination
- E — Eyes: sudden vision change in one or both eyes
- F — Face drooping: one side of face droops (ask to smile)
- A — Arm weakness: one arm drifts down when raised
- S — Speech difficulty: slurred, strange, or inability to speak
- T — Time: call emergency services IMMEDIATELY

Ischaemic Stroke Treatment
- IV tPA (alteplase) within 4.5 h of onset
- Mechanical thrombectomy within 24 h (large vessel occlusion)
- Aspirin 300 mg stat (if haemorrhage excluded)
- Blood pressure management
- Statins, antihypertensives for secondary prevention
- Anticoagulation for cardioembolic stroke (AF)

Haemorrhagic Stroke Treatment
- Control blood pressure urgently
- Reverse anticoagulation (Vitamin K, PCC, FFP)
- Surgical evacuation (select cases)
- ICP management (mannitol, head elevation)
- Neurosurgical consultation
- Seizure prophylaxis

CHAPTER 3

Respiratory Diseases

Asthma · COPD · Pneumonia · Pulmonary Embolism

3.1 Asthma

Asthma is a chronic inflammatory airway disease characterised by reversible airway obstruction, bronchial hyperresponsiveness, and remodelling. Triggers precipitate acute exacerbations (asthma attacks).

Common Triggers	Treatment — Stepwise (GINA)
• Allergens (pollen, dust mites, pet dander, mould) • Respiratory tract infections (viral) • Exercise (exercise-induced bronchospasm) • Cold air and air pollution • NSAIDs / Aspirin (in AERD) • Occupational exposures • Strong emotions / stress	• Step 1–2: SABA (Salbutamol) PRN ± low-dose ICS • Step 3: Low-dose ICS + LABA (Formoterol) • Step 4: Medium/high-dose ICS + LABA • Step 5: Add-on Tiotropium, Biologics (Omalizumab, Mepolizumab) • Acute exacerbation: SABA, O_2, systemic corticosteroids • Avoid triggers; annual influenza vaccine

3.2 Chronic Obstructive Pulmonary Disease (COPD)

COPD encompasses emphysema and chronic bronchitis, characterised by progressive, irreversible airflow limitation. Smoking accounts for ~80% of cases. $FEV_1/FVC < 0.7$ post-bronchodilator confirms diagnosis (GOLD spirometry).

Symptoms	Management
• Chronic productive cough ('smoker's cough') • Breathlessness (progressive dyspnoea) • Wheeze and chest tightness • Recurrent respiratory infections • Barrel chest (emphysema) • Cyanosis, cor pulmonale (advanced)	• Smoking cessation — single most effective intervention • SABA / LABA bronchodilators • LAMA (Tiotropium, Umeclidinium) • ICS + LABA combination (high exacerbation risk) • Pulmonary rehabilitation • Long-term O_2 therapy (if $PaO_2 < 55$ mmHg) • Annual flu + pneumococcal vaccines

3.3 Pneumonia

Pneumonia is an infection of the lung parenchyma caused by bacteria (Streptococcus pneumoniae, Haemophilus influenzae), viruses (influenza, SARS-CoV-2), fungi (Pneumocystis jirovecii in immunocompromised), or aspiration.

Symptoms	Treatment
• Fever with rigors and sweats • Productive cough (rust-coloured sputum in pneumococcal) • Pleuritic chest pain (sharp, worse on breathing) • Dyspnoea and tachypnoea • Decreased breath sounds / crackles • Confusion in elderly (atypical presentation)	• CAP (mild): Amoxicillin ± Clarithromycin (outpatient) • CAP (moderate-severe): IV Co-amoxiclav + Macrolide • Atypical CAP: Doxycycline or Macrolide • HAP/VAP: Piperacillin-Tazobactam ± Aminoglycoside • PCP (in HIV): Co-trimoxazole (high dose) + Steroids • O_2 supplementation; fluid management

CHAPTER 4

Metabolic & Endocrine Disorders

Diabetes · Thyroid · Obesity · Adrenal

4.1 Diabetes Mellitus

Feature	Type 1 DM	Type 2 DM	Gestational DM
Mechanism	Autoimmune beta-cell destruction	Insulin resistance + relative deficiency	Placental hormones cause IR
Onset	Childhood/adolescence	Adulthood (increasing in youth)	Pregnancy (2nd/3rd trimester)
BMI	Usually normal	Often overweight/obese	Variable
Treatment	Insulin (basal-bolus)	Lifestyle + oral agents + insulin if needed	Diet, exercise, insulin if needed
Key drugs	Insulin (all types)	Metformin, GLP-1 RA, SGLT-2i, DPP-4i, SU	Insulin, Metformin
Monitoring	HbA1c, CGM, glucose logs	HbA1c <7%, renal/eye/foot checks	Fasting & postprandial glucose

Diabetic Emergencies

• Hypoglycaemia (<70 mg/dL): Sweating, tremor, palpitations, confusion → 15g fast carbs (rule of 15)

• DKA (Type 1): High glucose, ketones, metabolic acidosis → IV fluids, insulin, K+ replacement

• HHS (Type 2): Extreme hyperglycaemia without ketosis → Aggressive rehydration, insulin drip

• Chronic complications: Retinopathy, nephropathy, neuropathy, macrovascular disease

4.2 Thyroid Disorders

Hypothyroidism
• Cause: Hashimoto's thyroiditis (autoimmune), iodine deficiency
• Symptoms: fatigue, weight gain, cold intolerance, constipation, bradycardia, depression
• Diagnosis: TSH elevated, free T4 low
• Treatment: Levothyroxine (T4 replacement) — lifelong
• Myxoedema coma: severe emergency — IV T4/T3 + steroids

Hyperthyroidism
• Cause: Graves' disease (TSI antibodies), toxic nodule
• Symptoms: weight loss, heat intolerance, palpitations, tremor, exophthalmos (Graves')
• Diagnosis: TSH low, free T4/T3 elevated
• Treatment: Carbimazole/PTU, radioiodine, thyroidectomy
• Thyroid storm: emergency — beta-blockers, PTU, iodine, steroids

CHAPTER 5

Neurological Disorders

Brain · Spine · Nerves · Mental-Motor Function

5.1 Alzheimer's Disease

Alzheimer's disease (AD) is the most common cause of dementia (~60–70%), characterised by accumulation of amyloid-beta plaques and neurofibrillary tau tangles. It is progressive and incurable; treatment is symptomatic.

Stages & Symptoms	Treatment
• Early: short-term memory loss, word-finding difficulty • Moderate: confusion, disorientation in familiar places • Late: loss of speech, inability to swallow, bedridden • Neuropsychiatric: agitation, wandering, sleep disturbance • Loss of ADLs (activities of daily living)	• Cholinesterase inhibitors (mild-moderate): Donepezil, Rivastigmine, Galantamine • NMDA antagonist (moderate-severe): Memantine • Lecanemab (anti-amyloid monoclonal Ab) — newer FDA-approved • Caregiver support, cognitive stimulation therapy • Safety measures (wander-guard, home modification) • Treat comorbidities: depression, sleep disorders

5.2 Parkinson's Disease

Parkinson's disease is a progressive neurodegenerative disorder caused by loss of dopaminergic neurons in the substantia nigra, leading to the cardinal motor features of TRAP.

Cardinal Features (TRAP)

• T — Tremor: resting 'pill-rolling' tremor (4–6 Hz), diminishes with movement

• R — Rigidity: cogwheel or lead-pipe rigidity

• A — Akinesia/Bradykinesia: slowness and poverty of movement (micrographia, hypomimia)

• P — Postural instability: falls, festinant gait, freezing episodes

Non-Motor Features	Treatment
• Anosmia (early sign, preclinical) • REM sleep behaviour disorder • Constipation and autonomic dysfunction • Depression and anxiety • Cognitive impairment / Parkinson's dementia • Orthostatic hypotension	• Levodopa + Carbidopa — most effective • Dopamine agonists (Pramipexole, Ropinirole) • MAO-B inhibitors (Selegiline, Rasagiline) • COMT inhibitors (Entacapone) • DBS (deep brain stimulation) — advanced disease • Physiotherapy, speech therapy, occupational therapy

5.3 Epilepsy

Epilepsy is defined by ≥2 unprovoked seizures >24 hours apart, or 1 unprovoked seizure with high recurrence risk. It affects ~50 million people worldwide. Seizure classification guides antiepileptic drug (AED) selection.

Seizure Types	First-Line AEDs
• Focal aware (simple partial): consciousness preserved • Focal impaired awareness (complex partial): impaired consciousness • Focal to bilateral tonic-clonic • Generalised tonic-clonic (grand mal) • Absence (petit mal): blank stare 5–10 s • Myoclonic: brief muscle jerks • Atonic: 'drop attacks'	• Focal: Levetiracetam, Lamotrigine, Carbamazepine • Generalised: Sodium Valproate, Lamotrigine, Levetiracetam • Absence: Ethosuximide, Valproate • Status epilepticus (emergency): • IV Lorazepam → IV Levetiracetam / Phenytoin • Refractory: IV Propofol or Midazolam infusion

CHAPTER 6

Gastrointestinal Diseases

Oesophagus · Stomach · Bowel · Liver

6.1 GERD (Gastro-Oesophageal Reflux Disease)

GERD occurs when stomach acid repeatedly flows back into the oesophagus, causing symptoms and potential mucosal injury. Untreated, it can progress to Barrett's oesophagus and oesophageal adenocarcinoma.

Symptoms	Treatment
• Heartburn (retrosternal burning) • Regurgitation of acid • Dysphagia (difficulty swallowing) • Chest pain (non-cardiac) • Chronic cough, hoarseness (atypical) • Dental erosions • Globus sensation ('lump in throat')	• Lifestyle: weight loss, elevate bed head, avoid late meals • Antacids: Gaviscon, Calcium carbonate (quick relief) • H2 blockers: Ranitidine, Famotidine • PPIs (first-line pharmacotherapy): Omeprazole, Pantoprazole • Nissen fundoplication (surgery) for refractory/severe cases • Upper GI endoscopy if red flags present

6.2 Inflammatory Bowel Disease (IBD)

Crohn's Disease	Ulcerative Colitis
• Transmural inflammation — any GI segment • Skip lesions, cobblestone mucosa • Perianal disease, fistulae, strictures • Malabsorption, weight loss • Extra-intestinal: uveitis, erythema nodosum • Rx: Steroids, Azathioprine, Methotrexate, Anti-TNF (Infliximab)	• Mucosal inflammation — rectum, extends proximally • Continuous involvement, no skip lesions • Bloody diarrhoea with mucus • Tenesmus, urgency, abdominal cramps • Risk of toxic megacolon (emergency) • Rx: 5-ASA (Mesalazine), Steroids, Biologics, Colectomy (refractory)

Note: Both Crohn's and UC carry increased colorectal cancer risk — surveillance colonoscopy recommended after 8–10 years of disease.

6.3 Liver Cirrhosis

Cirrhosis is end-stage liver fibrosis from chronic injury (alcohol, viral hepatitis, NAFLD/NASH). Irreversible scarring disrupts hepatic architecture, causing portal hypertension and liver failure.

Complications of Cirrhosis

• Portal hypertension → Oesophageal/gastric varices (risk of massive haemorrhage)

- Ascites (fluid in peritoneum) — spironolactone, furosemide; paracentesis if tense
- Hepatic encephalopathy — lactulose, rifaximin; low protein diet (outdated, avoid protein restriction)
- Spontaneous bacterial peritonitis (SBP) — IV cefotaxime; prophylactic norfloxacin
- Hepatorenal syndrome — vasoconstrictors (terlipressin) + albumin
- Hepatocellular carcinoma (HCC) — 6-monthly AFP + ultrasound surveillance

CHAPTER 7

Mental Health Disorders

Depression · Anxiety · Psychosis · PTSD

7.1 Major Depressive Disorder (MDD)

MDD is characterised by ≥2 weeks of persistent depressed mood or anhedonia with functional impairment. It is the leading cause of disability worldwide, affecting ~280 million people.

DSM-5 Diagnostic Criteria (≥5 symptoms, ≥2 weeks, including 1 of first 2)

- 1. Depressed mood most of the day, nearly every day
- 2. Anhedonia (loss of interest/pleasure in activities)
- 3. Significant weight change or appetite disturbance
- 4. Insomnia or hypersomnia
- 5. Psychomotor agitation or retardation
- 6. Fatigue or loss of energy
- 7. Feelings of worthlessness or excessive guilt
- 8. Difficulty concentrating or making decisions
- 9. Recurrent thoughts of death or suicidal ideation

Pharmacotherapy

- First-line: SSRIs (Fluoxetine, Sertraline, Escitalopram)
- SNRIs: Venlafaxine, Duloxetine
- TCAs: Amitriptyline (2nd line, pain comorbidity)
- MAOIs: Phenelzine (refractory, dietary restrictions)
- Mirtazapine (with insomnia/poor appetite)
- Treat for ≥6–12 months after remission

Psychotherapy

- CBT (Cognitive Behavioural Therapy) — first-line
- Interpersonal therapy (IPT)
- Behavioural activation
- Mindfulness-based cognitive therapy (MBCT) — relapse prevention
- ECT (electroconvulsive therapy) — severe/refractory/psychotic
- TMS (transcranial magnetic stimulation) — newer option

7.2 Anxiety Disorders

Disorder	Core Feature	First-Line Treatment
GAD	Excessive worry ≥6 months about multiple domains	CBT + SSRI/SNRI

Panic Disorder	Recurrent unexpected panic attacks with anticipatory anxiety	CBT, SSRI; Alprazolam (short-term)
Social Anxiety	Marked fear of social scrutiny/embarrassment	CBT, SSRI, SNRI
OCD	Obsessions + compulsions >1 h/day	CBT (ERP) + SSRI (high dose)
PTSD	Re-experiencing, avoidance, hyperarousal after trauma	Trauma-focused CBT, EMDR, Sertraline/Paroxetine
Specific Phobia	Irrational fear of specific object/situation	Exposure therapy

CHAPTER 8

Musculoskeletal Disorders

Joints · Bones · Muscles · Connective Tissue

8.1 Osteoarthritis (OA)

OA is the most common joint disease — a degenerative condition involving progressive cartilage loss, subchondral bone changes, and synovial inflammation. Most commonly affects knees, hips, hands, and spine.

Symptoms	Management
• Joint pain worsening with activity, relieved by rest • Morning stiffness <30 min (vs. >60 min in RA) • Crepitus on movement • Joint deformity (Heberden's/Bouchard's nodes — hands) • Decreased range of motion • No systemic features (differentiates from inflammatory arthritis)	• Non-pharmacological: exercise, weight loss, physiotherapy, walking aids • Topical NSAIDs (first-line pharmacological for knee/hand) • Oral paracetamol / NSAIDs (GI protection with PPI) • Intra-articular corticosteroids (flares) • Duloxetine (central sensitisation/chronic pain) • Total joint replacement (severe, refractory disease)

8.2 Rheumatoid Arthritis (RA)

RA is a chronic autoimmune systemic inflammatory disease primarily affecting synovial joints symmetrically. Anti-CCP antibodies and Rheumatoid Factor (RF) are diagnostic markers. Early aggressive treatment prevents erosive joint damage.

Clinical Features	Treatment (Treat-to-Target)
• Symmetrical small joint involvement (MCP, PIP, wrists) • Morning stiffness >1 hour • Rheumatoid nodules (over pressure points) • Systemic: fatigue, low-grade fever, weight loss • Extra-articular: pericarditis, scleritis, interstitial lung disease • Cervical spine instability (atlantoaxial subluxation)	• DMARDs (first-line): Methotrexate (anchor drug) • Hydroxychloroquine, Sulfasalazine, Leflunomide • Biologics: Anti-TNF (Etanercept, Adalimumab) • IL-6 inhibitors (Tocilizumab), B-cell depletion (Rituximab) • JAK inhibitors: Baricitinib, Upadacitinib • Bridging: short-course corticosteroids

8.3 Osteoporosis

Osteoporosis is defined by reduced bone mineral density (T-score ≤ -2.5 on DEXA scan), predisposing to fragility fractures. Most common sites: vertebrae, hip, and distal radius.

Risk Factors	Treatment
• Female sex, post-menopause • Age >65 • Family history of fragility fracture • Chronic corticosteroid use • Smoking, excessive alcohol • Malabsorption, coeliac disease • Rheumatoid arthritis, hypogonadism	• Calcium (1000–1200 mg/day) + Vitamin D (800–1000 IU/day) • Bisphosphonates: Alendronate, Zoledronic acid (first-line) • Denosumab (RANKL inhibitor) — subcutaneous injection • Teriparatide (PTH analogue) — severe osteoporosis • Romosozumab (sclerostin inhibitor) — high fracture risk • Weight-bearing exercise; fall prevention strategies

CHAPTER 9

Oncology — Cancer Overview

Types · Staging · Treatment Modalities · Palliative Care

Cancer is defined by uncontrolled cell proliferation, invasion of adjacent tissues, and potential distant metastasis. Hallmarks include: sustained proliferative signalling, evasion of apoptosis, replicative immortality, angiogenesis, and immune evasion.

Cancer Warning Signs (CAUTION)

- C — Change in bowel or bladder habits
- A — A sore that does not heal
- U — Unusual bleeding or discharge
- T — Thickening or lump in breast or elsewhere
- I — Indigestion or difficulty swallowing
- O — Obvious change in a wart or mole
- N — Nagging cough or hoarseness

Cancer	Key Risk Factors	Screening	Treatment Overview
Breast	BRCA1/2, oestrogen exposure, family hx	Mammography (40–74 yr)	Surgery, radiation, hormone therapy, HER2 biologics, chemo
Lung	Smoking (85%), radon, asbestos	Low-dose CT (heavy smokers 50–80 yr)	Surgery (early), chemo, targeted therapy (EGFR, ALK), immunotherapy
Colorectal	Diet, IBD, Lynch syndrome, polyps	Colonoscopy from 45 yr	Polypectomy (early), surgery, FOLFOX/FOLFIRI, bevacizumab
Prostate	Age, African ancestry, BRCA2	PSA + digital rectal exam	Active surveillance, radical prostatectomy, radiotherapy, ADT
Cervical	HPV 16/18, multiple partners, smoking	Pap smear + HPV test (21–65 yr)	HPV vaccine, LEEP, hysterectomy, chemo-radiation
Leukaemia	Radiation, Down syndrome, benzene, prior chemo	No routine screening	Chemotherapy, targeted TKIs (Imatinib for CML), stem cell transplant

Treatment Modalities

Curative / Primary Modalities	Supportive / Palliative
• Surgery: resection of primary tumour ± lymph nodes • Radiotherapy: external beam (EBRT), stereotactic (SBRT/SRS), brachytherapy • Chemotherapy: cytotoxic agents (DNA damage, mitotic arrest) • Targeted therapy: EGFR inhibitors, HER2 inhibitors, BRAF inhibitors • Immunotherapy: PD-1/PD-L1 checkpoint inhibitors (Pembrolizumab, Nivolumab) • CAR-T cell therapy: haematological malignancies	• Pain management (WHO analgesic ladder) • Antiemetics for chemotherapy-induced nausea • G-CSF for neutropenia prevention (Filgrastim) • Bisphosphonates for bone metastases • Psychological support and counselling • Hospice and end-of-life care planning

CHAPTER 10

Dermatological Conditions

Skin · Nails · Hair · Mucous Membranes

10.1 Eczema (Atopic Dermatitis)

Atopic dermatitis is a chronic relapsing pruritic inflammatory skin condition linked to skin barrier dysfunction (filaggrin mutation) and Th2-mediated immune response. Strongly associated with asthma and allergic rhinitis (atopic triad).

Features	Treatment
• Intense pruritus ('the itch that rashes') • Erythematous, weeping, crusted lesions (acute) • Lichenification, dry skin (chronic) • Distribution: flexural creases (cubital, popliteal) • Infants: face, scalp, extensor surfaces • IgE elevation, peripheral eosinophilia	• Emollients (moisturisers) — mainstay of daily care • Topical corticosteroids (mild-potent depending on site) • Topical calcineurin inhibitors (Tacrolimus, Pimecrolimus) • PDE4 inhibitors: Crisaborole • Biologics (moderate-severe): Dupilumab (IL-4/13 blockade) • JAK inhibitors: Abrocitinib, Upadacitinib • Trigger avoidance, antihistamines for itch

10.2 Psoriasis

Psoriasis is a chronic, immune-mediated hyperproliferative skin disorder with well-demarcated, erythematous, silvery-scaled plaques. Extracutaneous manifestations include psoriatic arthritis (up to 30%) and increased cardiovascular risk.

Clinical Types	Treatment
• Plaque psoriasis (most common, 85%) — silvery scales on extensor surfaces • Guttate psoriasis — drop-shaped lesions post-streptococcal infection • Pustular psoriasis — sterile pustules; can be life-threatening • Erythrodermic psoriasis — generalised erythema, thermoregulation failure • Nail psoriasis — pitting, onycholysis, oil-drop sign	• Mild: topical steroids, Vitamin D analogues (Calcipotriol), coal tar • Moderate: Phototherapy (NB-UVB) • Systemic: Methotrexate, Acitretin, Ciclosporin • Biologics: Anti-TNF, IL-17 inhibitors (Secukinumab, Ixekizumab) • IL-23 inhibitors (Risankizumab, Guselkumab) — highly effective • PDE4 inhibitor: Apremilast (oral)

10.3 Melanoma (Skin Cancer)

Melanoma arises from melanocytes and is the most dangerous form of skin cancer due to its high metastatic potential. Early detection is critical — 5-year survival >98% (stage I) vs <20% (stage IV).

ABCDE Warning Signs

• A — Asymmetry: one half doesn't match the other

• B — Border: irregular, ragged, or blurred borders

• C — Colour: variation in colour (tan, brown, black, red, white, blue)

• D — Diameter: >6 mm (size of a pencil eraser)

• E — Evolving: any change in size, shape, colour, or new symptoms (bleeding, itching)

CHAPTER 11

Urological & Renal Diseases

Kidneys · Urinary Tract · Prostate

11.1 Chronic Kidney Disease (CKD)

CKD is defined as kidney damage or eGFR <60 mL/min/1.73m² for ≥3 months. Staging by eGFR (G1–G5) and albuminuria (A1–A3) guides prognosis and management. Main causes: Diabetes (40%), Hypertension (25%), Glomerulonephritis.

CKD Stages & eGFR

- G1: eGFR ≥90 mL/min — normal/high; monitor if albuminuria present
- G2: eGFR 60–89 mL/min — mildly decreased
- G3a/G3b: eGFR 30–59 mL/min — mild to moderately decreased; nephrology referral
- G4: eGFR 15–29 mL/min — severely decreased; prepare for RRT
- G5: eGFR <15 mL/min — kidney failure; dialysis or transplant

Complications

- Anaemia (erythropoietin deficiency) — ESA therapy
- Metabolic acidosis — sodium bicarbonate
- Hyperphosphataemia — phosphate binders
- Hyperkalaemia — dietary restriction, Patiromer
- Hypertension — ACEi/ARB (renoprotective)
- Renal osteodystrophy — calcitriol, cinacalcet
- Uraemic symptoms (nausea, pruritis, encephalopathy)

Renal Replacement Therapy (G5)

- Haemodialysis (HD): 3x/week, 4h sessions
- Peritoneal dialysis (PD): home-based, continuous
- Renal transplantation: best long-term outcome
- Conservative management (elderly/comorbid)
- Access: arteriovenous fistula (HD), Tenckhoff catheter (PD)
- Immunosuppression post-transplant: Tacrolimus, MMF, Prednisolone

11.2 Urinary Tract Infections (UTI)

Uncomplicated UTI (Lower)

- Dysuria, frequency, urgency
- Suprapubic pain, haematuria
- Commonest pathogen: E. coli (80%)
- Diagnosis: MSU dipstick / culture
- Treatment: Nitrofurantoin 3–5d or Trimethoprim 7d
- Cephalexin / Fosfomycin — alternatives

Complicated UTI / Pyelonephritis

- Fever, rigors, loin pain, CVA tenderness
- Nausea, vomiting, haematuria
- Risk factors: pregnancy, catheter, obstruction, immunosuppression
- Oral: Ciprofloxacin 7–14d (if sensitivities permit)
- IV: Ceftriaxone or Piperacillin-Tazobactam (severe/sepsis)
- Blood and urine cultures before starting antibiotics

CHAPTER 12

Pharmacology & Medical Insights

Drug Mechanisms · Safety · Vaccines · Pain Management

12.1 Key Drug Classes & Mechanisms of Action

Drug Class	Examples	Mechanism	Key Indications
Beta-blockers	Metoprolol, Carvedilol	Block β1/β2 adrenoceptors → ↓HR, ↓BP, ↓inotropy	Hypertension, Heart failure, Angina, Post-MI
ACE inhibitors	Enalapril, Ramipril	Inhibit ACE → ↓Angiotensin II → vasodilation, ↓aldosterone	HTN, Heart failure, DM nephropathy, Post-MI
Statins	Atorvastatin, Rosuvastatin	Inhibit HMG-CoA reductase → ↓LDL synthesis	Hyperlipidaemia, CVD prevention
SSRIs	Fluoxetine, Sertraline	Block serotonin reuptake transporter (SERT) → ↑synaptic 5-HT	Depression, Anxiety, OCD, PTSD
Benzodiazepines	Diazepam, Lorazepam	Positive allosteric modulator of GABA-A receptor → CNS depression	Anxiety, Seizures, Alcohol withdrawal
Opioids	Morphine, Oxycodone, Fentanyl	Agonist at μ (mu) opioid receptors → analgesia, euphoria, respiratory depression	Moderate-severe pain, Palliative care
Proton Pump Inhibitors	Omeprazole, Pantoprazole	Irreversibly inhibit H+/K+-ATPase → ↓gastric acid	GERD, PUD, H. pylori eradication
Metformin	Metformin	Activates AMPK → ↓hepatic gluconeogenesis, ↑insulin sensitivity	Type 2 Diabetes (first-line)
GLP-1 Receptor Agonists	Semaglutide, Liraglutide	Mimic GLP-1 → ↑insulin (glucose-dependent), ↓glucagon, ↓appetite	Type 2 DM, Obesity, CVD risk reduction
Warfarin	Warfarin	Inhibits Vitamin K epoxide reductase → ↓clotting factors II, VII, IX, X	AF (stroke prevention), VTE, mechanical heart valves
DOACs	Rivaroxaban, Apixaban, Dabigatran	Direct Xa inhibitors (Rivaroxaban/Apixaban) or direct thrombin inhibitor (Dabigatran)	AF, VTE treatment/prevention
Antibiotics — Penicillins	Amoxicillin, Flucloxacillin	Inhibit bacterial cell wall synthesis (PBP binding)	Broad bacterial infections; Flucloxacillin for Staph
Macrolides	Azithromycin, Clarithromycin	Bind 50S ribosomal subunit → ↓protein synthesis (bacteriostatic)	Atypical pneumonia, STIs, H. pylori
Fluoroquinolones	Ciprofloxacin, Levofloxacin	Inhibit DNA gyrase and topoisomerase IV → ↓DNA replication	UTI, pneumonia, TB (2nd line)

12.2 Antibiotic Stewardship Principles

Core Principles	Antimicrobial Resistance (AMR)
• Right drug — narrow-spectrum preferred when culture available • Right dose — weight-based, renal dose adjustment • Right route — oral preferred over IV when feasible • Right duration — shortest effective course • Review at 48–72 h: de-escalate if sensitivities allow • Avoid antibiotics in viral infections (e.g., common cold)	• ESKAPE pathogens: Enterococcus, Staph aureus (MRSA), Klebsiella, Acinetobacter, Pseudomonas, Enterobacteriaceae • MRSA: Treat with Vancomycin or Linezolid • ESBL producers: Carbapenems (Meropenem) • Carbapenem-resistant: Colistin, Ceftazidime-avibactam • Culture & sensitivity must guide definitive therapy • Infection control: hand hygiene, contact precautions

12.3 Vaccine Immunology & Schedule Highlights

Vaccines harness the immunological memory of adaptive immunity. After antigen exposure, B-cell differentiation produces antibodies and long-lived plasma cells; T-cell priming creates cytotoxic CD8+ and helper CD4+ memory cells.

Key Vaccines by Type

• Live attenuated: MMR, BCG, Varicella, Yellow Fever, Oral Polio — contraindicated in immunocompromised

• Inactivated/killed: IPV (polio), Hepatitis A, Influenza (injectable), Rabies

• Subunit/recombinant: Hepatitis B, HPV (Gardasil-9), Pertussis component, Shingrix

• Toxoid: Tetanus, Diphtheria — antibodies against bacterial toxins

• mRNA vaccines: COVID-19 (Pfizer-BioNTech, Moderna) — novel platform, no DNA alteration

• Conjugate vaccines: Hib, PCV13, MenACWY — T-cell-dependent response in infants

• Viral vector: COVID-19 (AstraZeneca, Johnson & Johnson), Ebola (Ervebo)

12.4 WHO Analgesic Ladder — Pain Management

Step	Pain Level	Analgesics
Step 1	Mild pain (NRS 1–3)	Non-opioids: Paracetamol, NSAIDs (Ibuprofen, Naproxen) ± Adjuvants
Step 2	Moderate pain (NRS 4–6)	Weak opioids: Codeine, Tramadol, Dihydrocodeine + Step 1 drugs ± Adjuvants
Step 3	Severe pain (NRS 7–10)	Strong opioids: Morphine, Oxycodone, Fentanyl, Hydromorphone + Step 1 drugs ± Adjuvants

12.5 Dangerous Drug Interactions — Quick Reference

Drug A	Drug B	Interaction	Clinical Effect
Warfarin	Aspirin / NSAIDs	Pharmacodynamic	↑Bleeding risk — haemorrhage
SSRIs	MAOIs	Serotonin syndrome	Potentially fatal: hyperthermia, myoclonus, agitation
Statins	Amiodarone / Diltiazem	CYP3A4 inhibition	↑Statin levels → rhabdomyolysis
Metformin	IV Contrast dye	Renal impairment risk	↑Lactic acidosis — hold 48 h post-contrast
ACE inhibitors	Potassium-sparing diuretics	Pharmacodynamic	Hyperkalaemia — cardiac arrhythmia
Azithromycin	QT-prolonging drugs	QT prolongation	Torsades de pointes / sudden cardiac death
Fluconazole	Warfarin	CYP2C9 inhibition	↑Warfarin → ↑INR → bleeding
Lithium	NSAIDs / Thiazides	↓Renal Li excretion	Lithium toxicity — neurological emergency

12.6 Medical Abbreviations & Vital Sign Reference

Normal Adult Vital Signs

- Temperature: 36.1–37.2 °C (97–99 °F)
- Heart rate: 60–100 bpm
- Respiratory rate: 12–20 breaths/min
- Blood pressure: <120/80 mmHg (optimal)
- SpO2: ≥95% on room air
- Blood glucose (fasting): 70–99 mg/dL (3.9–5.5 mmol/L)
- GCS: 15/15 (normal consciousness)

Common Lab Reference Ranges

- Haemoglobin: M 130–175, F 115–155 g/L
- WBC: 4.0–11.0 × 10^9/L
- Platelets: 150–400 × 10^9/L
- Creatinine: M 62–106, F 44–80 μmol/L
- eGFR: ≥60 mL/min/1.73m² (normal)
- Sodium: 135–145 mmol/L
- Potassium: 3.5–5.0 mmol/L
- TSH: 0.4–4.0 mIU/L
- HbA1c: <42 mmol/mol (<6.0%) — non-diabetic

This Comprehensive Medical Reference Guide was generated for use as a structured knowledge base to power Healthcare AI/RAG systems. All information reflects evidence-based clinical guidelines current to 2024–2025. This document should not replace professional medical judgement. Content covers: Infectious Diseases · Cardiovascular · Respiratory · Metabolic · Neurological · GI · Mental Health · Musculoskeletal · Oncology · Dermatology · Renal · Pharmacology.